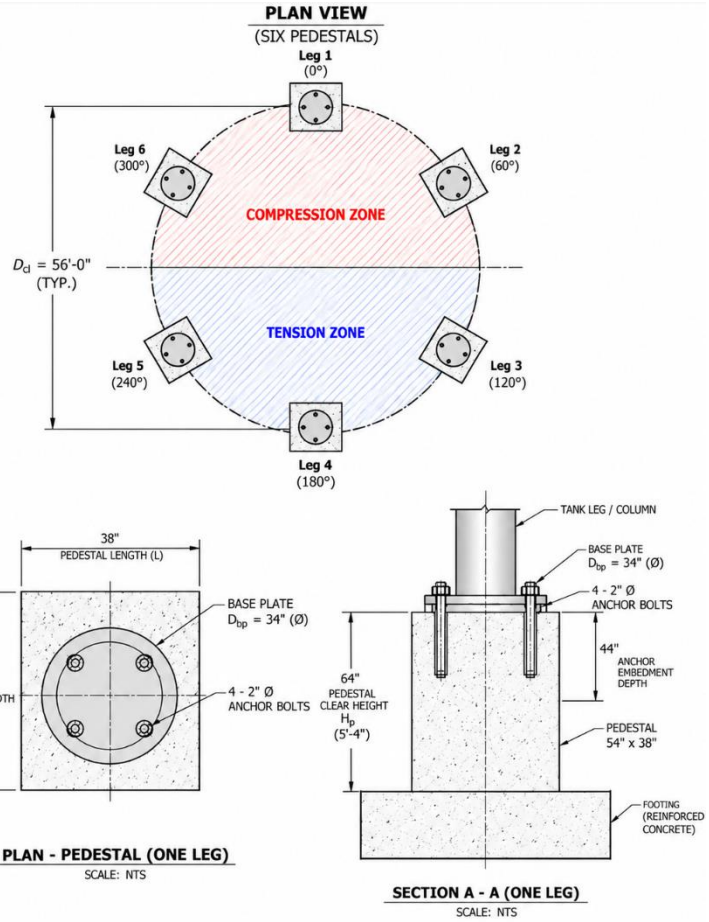
3: RINGWALL Foundation Design:

Design Parameter	Symbol	Value	Source
Total factored axial load	P_u	**7550.979**	Main Tank Design (Predefined)
Governing factored overturning moment	M_u	**13656.1295 kip-ft**	Main Tank Design (Predefined)
Total factored horizontal shear	V_u	**11.5.783 kips**	Main Tank Design (Predefined)
Number of legs	N_{leg}	**6**	User Input
Diameter circle representing axisymmetric spacing between legs/columns	D_{cl}	**56'-00"**	Main Tank Geometry (Predefined)
Pedestal size	$B \times L$	**54 in × 38 in**	User Input
Base plate size	D_{bp}	**34 in**	User Input
Concrete compressive strength	f_c'	**4.0 ksi**	User Input / ACI 318-19
Reinforcing steel yield strength	f_y	**36 ksi**	User Input / ACI 318-19
Anchor bolts	$N_b \times d_b$	**4 × 2 in**	Designed in Anchor Bolts segment of this software
Anchor embedment depth	h_{ef}	**44 in**	Designed in Anchor Bolts segment of this software
Pedestal clear height used	H_p	**≈ 5'-4" = 64 in**	User Input
Concrete unit weight	γ_c	**150 pcf**	User Input



Step 1-2: ACI 318-19 slenderness check

ACI 318-19 6.2.5.1 permits slenderness effects to be neglected for a column not braced against sidesway when:

For a square section:

$$r = \sqrt{\frac{I}{A}}$$

$$I_x = \frac{bh^3}{12} = \frac{54(38)^3}{12} = 246924 \text{ in}^4$$
$$I_y = \frac{hb^3}{12} = \frac{38(54)^3}{12} = 498636 \text{ in}^4$$

And:

$$A_g = 54(38)$$
$$A_g = 2052 \text{ in}^2$$

The minimum radius of gyration governs:

$$r_{min} = \sqrt{\frac{246924}{2052}}$$

r = 10.970 inch

Check A — Pedestal classification

$$\frac{\ell_u}{b_{min}} \leq 3$$
$$1.648 < 3$$

Result: PASS — geometrically satisfies the pedestal height-to-dimension criterion.

Check B — Structural Slenderness

Using the conservative $k = 2.0$:

$$\frac{k\ell_u}{r} = \frac{2(64)}{10.970} \leq 22$$

$$11.67 < 22$$

Therefore, second-order slenderness effects can be neglected.

Step 3: Gravity axial load per leg

For four equally spaced legs:

$$P_{gravity} = \frac{P_u}{N_{leg}}$$
$$P_g = \frac{7550.979}{4}$$
$$\boldsymbol{P_g = 1258.497 \text{ kip/leg}}$$

The pedestal self-weight can also be included.

For approximately 7'-5" pedestal height:

γ_c=150 pcf

$$W_p = (54/12)(38/12)(64/12)(0.150)/1000 \text{ kips}$$
$$W_p \approx 11.400 \text{ kips}$$

Factored:

$$1.2W_p = 13.680 \text{ kips}$$

Therefore, conservatively:

$$P_{g,total} = 1258.497 + 13.680$$

$$\boldsymbol{P_{g,total} = 1272.177 \text{ kips/leg}}$$

Step 4: Pedestal Concrete Bearing Check

Base plate:

$$A_1 = \frac{\pi(34)^2}{4} = 907.920 \text{ in}^2$$

Pedestal:

$$A_2 = 54(38) = 2052 \text{ in}^2$$

Bearing enhancement factor:

$$\sqrt{\frac{A_2}{A_1}} = \sqrt{\frac{2052}{907.920}} = 1.503$$

Using the concrete bearing-strength relationship: $\phi P_n = \phi(0.85 f_c' A_1)\sqrt{\frac{A_2}{A_1}}$

Taking:

Transverse Pedestal Reinforcement: if

1. Spiral = 0.75
2. Ties = 0.65

$$\phi P_n = 0.65(0.85)(4)(907.920)(1.503)$$
$$\boldsymbol{\phi P_n \approx 3016.510 \text{ kips}}$$

Demand:

$$P_u = \mathbf{792.336} \text{ kip}$$
$$\frac{\boldsymbol{Pmax}}{\boldsymbol{\phi P_n}} = \frac{\mathbf{1272.177}}{\mathbf{3016.510}} = 0.477$$

Therefore:

$0.477 < 1.00$……. **CONCRETE BEARING — PASS**

Demand/capacity ratio:……. $D/C \approx 0.262$

So, the 54 × 38 in pedestal is more than adequate for bearing.

Step 5: Pedestal longitudinal Reinforcement

Based ont the drawing* 18 #8 *vertical bars

Area of one #6:

$$A_b = 0.79 \text{ in}^2$$

$$A_s = 18(0.79)$$

$$A_s = 14.22 \text{ in}^2$$

$$A_g = 54(38) = 2052 \text{ in}^2$$

$$\rho_g = \frac{A_s}{A_g} = \frac{14.22}{2{,}052} = 0.00693$$

$$\rho_g = 0.693\%$$

$$A_{s,min} = 0.005A_g$$

ACI 318-19 Section 16.3.4.1 specifies the $0.005A_g$ minimum for connections between a cast-in-place column or pedestal and its foundation.

As,min=0.001(2052)

$$A_{s,min} = 10.26\ in^2$$

$$A_{s,\min} = 0.005(2{,}052) = 10.26\text{in}^2$$

Compare:

$$\mathbf{14.22 > 10.26....Pass}$$

Step 6: Axial Compression Capacity

$f_c' = 4.0\text{ksi}$

$f_y = 36\text{ksi}.$

For the illustrative concentric nominal axial strength:

$$P_o = 0.85f_c'(A_g - A_s) + f_y A_s$$
$$P_o = 0.85(4.0)(2{,}052 - 14.22) + 36(14.22)$$
$$P_o = 6928.452 + 511.92$$
$$P_o = 7{,}440.372 \text{ kips}$$

For a tied member, using the illustrative maximum axial-strength limit of $0.80P_o$ and $\phi = 0.65$:

$$\phi P_{n,\max} = 0.65(0.80)(7{,}440.372)$$
$$\phi P_{n,\max} = 3{,}869.0\text{kips}$$

Using $P_\text{u} = 1272.177\text{Kips}$:

$$D/C = \frac{1272.177}{3{,}762.2} = 0.339$$

Step 7: Pedestal Tensile Capacity Check

Using the foundation drawing's 18 #8 vertical bars:

$$A_s = 14.22\text{in}^2$$

Using your specified reinforcing steel yield strength, $f_y = 36\text{ksi}$, and assuming $\phi = 0.90$:

$$\phi T_n = \phi A_s f_y$$
$$\phi T_n = 0.90(14.22)(36)$$
$$\phi T_n = 460.728 \text{ kips}$$

Calculate the demand/capacity ratio:

Because the minimum force remains compressive, the net axial tensile demand is:

$$P_{u,\min} = 1109.604 \text{ kips (compression)}$$

$$D/C = \frac{T_u}{\phi T_n} = \frac{0}{460.728}$$
$$D/C = 0$$

No net axial uplift occurs under the assumed load combination.

Step 8: Pedestal *P-M* Interaction Check

Parameter	Value
Section width	54 in
Section depth	38 in
Concrete compressive strength	4.0 ksi
Reinforcing steel yield strength	36 ksi
Gross concrete area	2,052in^2
Longitudinal reinforcement	18 #8 bars
Total longitudinal steel area	14.22in^2

Design section properties

Steel yield strain

$E_s = 29{,}000\text{ksi}$

$$\epsilon_y = \frac{f_y}{E_s} = \frac{36}{29{,}000}$$
$$\epsilon_y = 0.001241$$

Balanced neutral-axis depth

The balanced neutral-axis depth

$$c_b = d\frac{0.003}{0.003 + \epsilon_y}$$

Substituting the steel yield strain:

$$c_b = d\frac{0.003}{0.003 + 0.001241}$$
$$c_b = 0.7073d$$

Equivalent compression-block depth

For $f_c' = 4.0$ksi, the applicable ACl stress-block factor must be confirmed for the governing code provisions. If $\beta_1 = 0.85$ applies:

$$a_b = \beta_1 c_b$$
$$a_b = 0.85(0.7073d)$$
$$a_b = 0.6012d$$

Step 9: Pedestal Tie Design

9.1 Tie diameter

ACI 318-19 Section 25.7.2.2 specifies the minimum tie-bar diameter based on longitudinal bar size:

- No. 3 ties for No. 10 or smaller longitudinal bars.
- No. 4 ties for No. 11 or larger longitudinal bars, or bundled longitudinal bars.

The foundation drawing specifies 14 No. 8 vertical bars. Therefore:

$$d_{bt} \geq \text{ No. } 3$$

For No. 3 ties:

$$d_{bt} = 0.375\text{in}$$

$$s_{\max} = \min(16d_b, 48d_{bt}, b_{\min})$$

For #8 bars, d_b=1.00in:

$$d_b = 1.00\text{in}$$

For the minimum No. 3 tie:

$$d_{bt} = 0.375 \text{ in}$$

The minimum pedestal dimension is:

$$b_{\min} = 38 \text{ in}$$

Calculate each limit:

$16d_b = 16(1.00)=16$inch

$48d_b = 48(0.375)=18$inch

$$b_{\min} = 38 \text{ in}$$

$$s_{\max} = \min(16,\ 18,\ 38)$$

$$s_{\max} = 16 \text{ in}$$

Adopt tie spacing

A preliminary spacing selection is:

$$s = 12 \text{ in}$$

$$12 < 16 \text{ in}$$

Thus, the selected spacing satisfies the calculated maximum-spacing limit.

Step 10: Pedestal-to-Footing Development

ACI 318-19 Section 25.4 provides the development and anchorage requirements for reinforcing bars. Because the foundation drawing specifies hooked vertical bars, the final calculation must use the applicable hooked-bar provisions if standard hooks are actually provided.

The following calculation documents the straight-bar development-length expression from your supplied calculation. It is not a substitute for the required hooked-bar check.

parameters

Parameter	**Symbol**	**Value**
Concrete compressive strength	f'c	4.0 ksi
Reinforcing steel yield strength	fy	60 ksi, as assumed in the supplied calculation
Lightweight concrete factor	λ	1.0
Top-bar factor	ψ_t	1.0, if applicable
Coating factor	ψ_t	1.0, if applicable
Bar-size factor	ψ_t	1.0 for No. 8 bars

Steel-grade factor	ψ_t	1.0 for Grade 60
Assumed confinement ratio	cb+K_{tr}/db	2.5
Bar diameter	d_b	1.00 in for a No. 8 bar

Calculate straight-bar development length

$$l_d = \frac{3}{40}\frac{f_y}{\lambda\sqrt{f'_c}}\frac{\psi_t\psi_e\psi_s\psi_g}{(c_b + K_{tr})/d_b}d_b$$

$$l_d = \frac{3}{40}\frac{36{,}000}{1.0\sqrt{4{,}000}}\frac{1.0(1.0)(1.0)(1.0)}{2.5}(1.00)$$

$$l_d = 17.08 \text{ in}$$

$$l_{\text{adopted}} = 18 \text{ in}$$

$$18 > 17.08 \text{ in}$$

Preliminary straight-bar length check: PASS